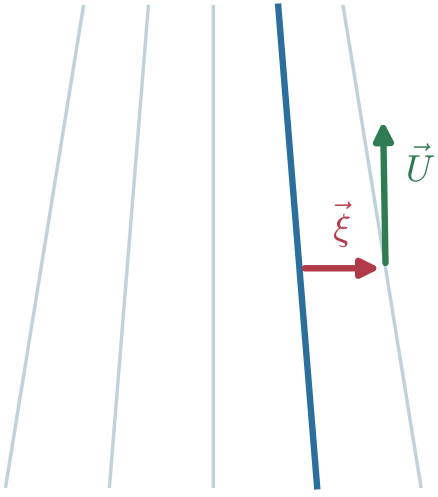
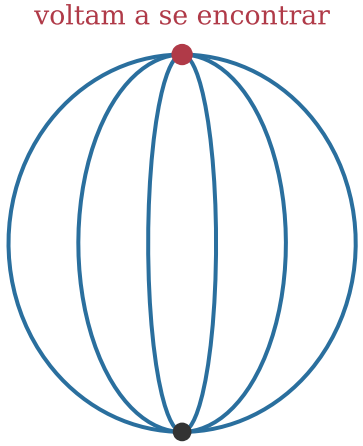
(a) a família



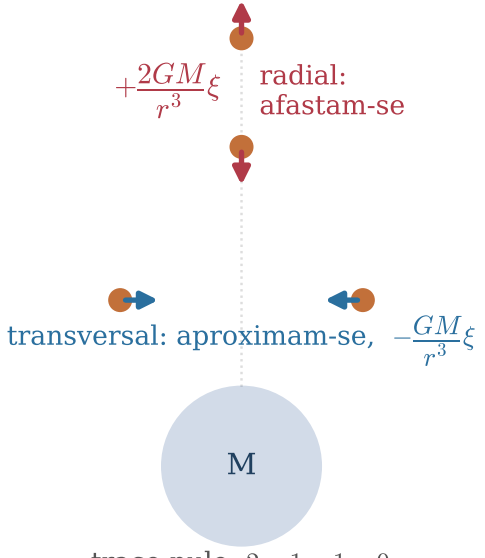
$\nabla_{\vec{U}}\vec{U}=0, \quad [\vec{U},\vec{\xi}]=0$

(b) esfera: $\ddot{\xi}=-\xi/a^2$



curvatura positiva foca

(c) e isso é a maré



traço nulo: $2-1-1=0$